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Metabasalts of the Harbour Main Group, eastern Newfoundland: a paleomagnetic study

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Abstract

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The Avalon peninsula of Newfoundland is distinguished from the central part of that island by its large proportion of Late Precambrian rocks. The study area is located in the Colliers Bay and Bauline areas at longitude 53° W and latitude 47° 30′N. Fifty-two oriented samples (140 specimen) were collected on ten different sites in the Harbour Main Group volcanics which are older than 610 Ma and younger than 770 Ma. Remanent magnetization and magnetic susceptibility were measured. Both alternating field and thermal demagnetizations were used to characterize the stability of magnetizations. Four components of magnetization (A, B, C, D) were isolated and their mean directions (317, −22; 175, +34; 190, −27; 266, −37) are corresponding to paleopoles at 20° N, 172° E; 24° N, 132° E; 56° N, 109° E; and 18° S, 205° E, respectively. Component D is encountered on two sites only and is largely overprinted by the present Earth's field. Component C is obtained in seven sites, is characterized by a significantly negative fold test and is thus presumed to be secondary. Components A and B are found on most sites and are also characterized by negative fold tests but the test is more significant for B than A. In addition, the larger unblocking temperatures and median destructive fields of component A suggest that it is older than B. A comparison of the paleopoles obtained with the apparent polar wander path for North America indicates that the age of acquisition of magnetization for C is Carbo-Permian and for B it is Upper Devonian; the age of D remains unresolved. The pole of component A obtained in this study lies close to that of component A obtained in a previous study on the Finn Hill ignimbrites; it is located within the Devonian–Late Hadrynian interval. It is tentatively interpreted as the signature of the Cadomian orogenic event that occurred at approximately 600 Ma.

Introduction

The Avalon peninsula (Fig. 1) is distinguished from central Newfoundland by its large proportion of Late Precambrian rocks (Malpas et al., 1974). These rocks were labelled "Avalonian" according to the name of the peninsula where they are typically exposed (Poole, 1967). The Avalonian rocks are both sedimentary and volcanic in composition and considerably younger than rocks of the Canadian Shield. No sedimentary rocks younger than Early Devonian are preserved in the Avalonian peninsula (Strong et al., 1980). Systematic mapping programs of the Avalon Zone were first undertaken by Rose (1952) in the Torbay–Bauline area (Fig. 1), and Hutchinson (1953) and McCartney (1956, 1957) to the south of this area. The most recent Precambrian stratigraphy of the Avalonian rocks was established by King (1979) and Strong (1979).

Precambrian rocks similar to the ones in the Avalon zone occur throughout the Appalachian

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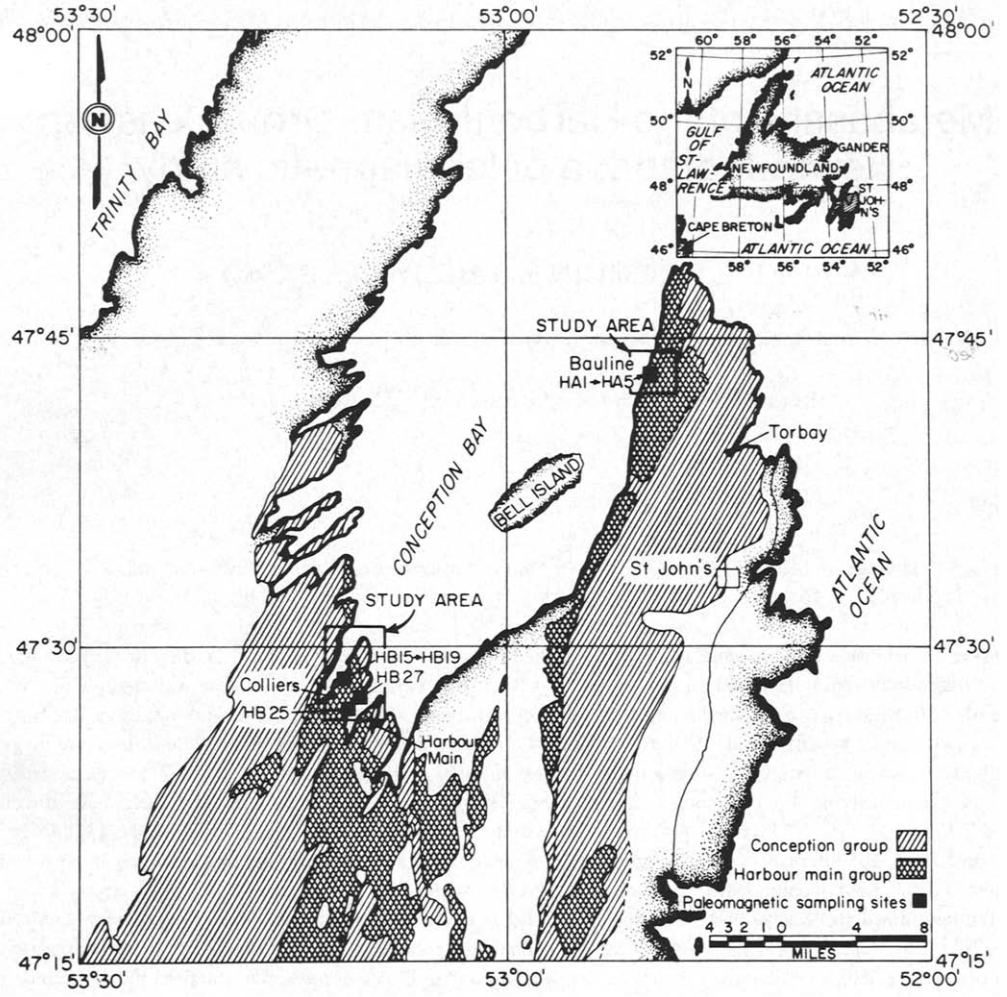


Fig. 1. General geology of the northern Avalon sector of Newfoundland and location of the study areas (Colliers and Bauline localities).

orogen in Newfoundland, Nova Scotia, New Brunswick, eastern Massachusetts, southeastern United States (Carolina Slate Belt; Schenk, 1971, 1978; Rankin, 1975; Williams, 1979) and on the eastern side of the Atlantic (Wales, Brittany, Iberian Peninsula, Czechoslovakia, Turkey as well as northeast Africa).

The Avalon Peninsula was effected by two main orogenies: a Late Precambrian Avalonian orogeny (Lilly, 1966; Hughes, 1970) and the Acadian orogeny of Middle to Late Devonian age (Schenk, 1978; Strong, 1979). Different tectonic histories have been proposed for the origin of the Avalon mini-continent. One of the tectonic models consists of volcanic islands (Hughes and Brückner, 1972) and ensialic island arc (Rast et al., 1976). A second model is related to subduction and Basin and Range-type rifting (Papezik, 1972; Schenk, 1978). A third tectonic model taking into account continental extension and rifting was advanced by Rankin (1976). In more recent models, the Avalon zone is considered as a terrane exotic to North America and a suspect terrane (Williams and Hatcher, 1983). From the geological standpoint, the Upper Proterozoic rock sequences of the Avalon zone of eastern Newfoundland are completely different from the Proterozoic rocks of the eastern North America (NA) craton occurring in western Newfoundland (Williams, 1978, 1979). If the Avalon plate is not of North American origin, where does it come from? And how did it get there? Geological comparisons of the Avalon Zone

with analogues of the southern British Isles (Rast et al., 1976) and North Africa (Schenk, 1971) are mainly based on the concept of orogenic belt: Monian in Britain, Cadomian in Europe and Pan-African in North Africa (Rast and Skehan, 1983). A comparison of the paleomagnetic signatures from the North African, Avalonian and Northwest European blocks for the Upper Proterozoic time interval is one of the possible tools to test the validity of the hypotheses put forward relative to the origin of the Avalon zone and its relative motion with respect to the adjacent plates (i.e. Laurentia, Gondwana and eventually Baltica). In addition, Precambrian paleomagnetism is poorly understood and this paper adds to the existing data set.

Geology

In the Colliers-Harbour Main and Bauline map-area (Fig. 1), the Late Precambrian rocks of the Avalon zone are subdivided into three groups: (1) Harbour Main Group volcanic rocks, (2) Conception Group composed of volcaniclastics, and (3) an upper sedimentary Group (Cabot Group) (Rose, 1952) which were renamed the St. John's and Signal Hill Groups by King (1979). North of Colliers, these groups are overlain by Lower Paleozoic sedimentary units. The basic flows of group 1 investigated in this study are described as alkali basalts by Papezik (1970, 1972). The mafic volcanic pile is weakly metamorphosed to the prehnite–pumpellyite subfacies of the subgreenschist facies (Nixon and Papezik, 1979). This unusually low metamorphic grade encountered in Precambrian rock units makes these geological formations particularly attractive for paleomagnetic investigations.

According to Nixon (1974) and Nixon and Papezik (1979), the mafic flows (basalts) sequence is steeply dipping to the west and is conformable with the underlying ash-flow tuffs (Bacon Cove, Weaver Hill and Finn Hill sequences). Individual flows are about 5 m thick and the total thickness is at least 600–800 m. The basalts consist of albitized plagioclase crystals, unaltered calcic augite, olivine more or less altered to serpentine, chlorite and hematite and smaller amounts of epidote, calcite, prehnite, pumpellyite and potash feldspar. The depositional environment is typically subaerial. The primary textures of these rocks are relatively well preserved and a lack of penetrative deformation is also evident (Nixon and Papezik, 1979). Consequently, some of these flows appear to be relatively pristine.

Age

In the light of world-wide correlation of fossils, tillites and rock units, Anderson (1972) suggested an age range of 800–600 Ma for the Conception Group which overlies the Harbour Main volcanic group. Rast et al. (1976) and Strong (1979) correlated the Love Cove Group volcanics from the Burin peninsula of southeastern Newfoundland with the Harbour Main Group volcanics. The Bell Bay volcanics which are younger than the Love Cove Group (Strong, 1979) have a $^{40}Ar/^{39}Ar$ age of 660 ± 20 Ma (Stukas, 1977). In establishing the relationships between Precambrian rocks of the Avalon platform in New Brunswick, the northeast Appalachians and the British Isles, Rast et al. (1976) assigned an age range of 750–650 Ma to the Coldbrook Group of New Brunswick which they correlated with the Love Cove and Harbour Main Groups. McCartney (1967) obtained a Rb–Sr age of 609 ± 11 Ma ($^{87}Rb = 1.34 \times 10^{-11}$ yrs) for the Holyrood granite which intruded this group. Thus, the Harbour Main (HMG) volcanics are older than 610 and younger than 762 Ma which is the age of the mafic volcanics in the Burin Group (Krogh et al., 1983), and which are definitely older than any strata on the Avalon peninsula (O'Brien and King, 1982); its more probable age is 675 Ma (Rast and Skehan, 1983). Consequently, King (1979) suggested that the Harbour Main Group is related in time to the Vendian Epoch of northern Eurasia. Recently, Krogh et al. (1983) obtained an age (U/Pb) of 621 ± 2 Ma for the Holyrood pluton intruding the HMG volcanics, thus providing an upper limit on the age.

Paleomagnetic sampling

Fifty two (52) oriented hand samples (140 specimens) were collected on ten different sites in

the Colliers Bay and Bauline areas (Fig. 1). The sample localities were chosen by reference to available geological maps (Rose, 1952; Nixon, 1975; Nixon and Papezik, 1979). Great care was taken to avoid the surficially altered outcrops. Orientations were obtained with a Brunton compass and some tests were done with a solar compass to detect eventual deflection of magnetic compass by the basalts. In all cases, the tests were negative. The samples consisted of green, grayish green or reddish-gray basaltic flows (Fig. 2). The red and black flows were not sampled because of their occasional excessive alteration. The oriented samples were cored in the laboratory and subsequently measured.

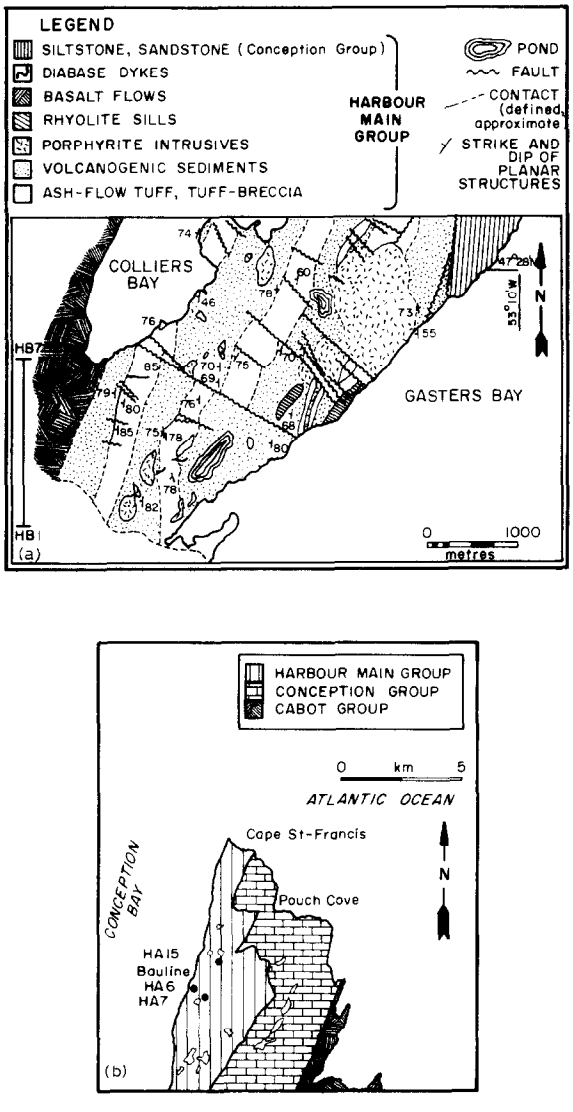


Fig. 2. Detailed geology of the study areas (after Rose, 1952, and Nixon and Papezik, 1979) with the distribution of the paleomagnetic sampling sites (HB1–HB7, HA6, HA7 and HA15).

Magnetization measurements

The direction and intensity of natural remanent magnetization (NRM) were measured with a Schonstedt digital spinner magnetometer (Model DSM-1). Room-temperature magnetic susceptibilities, k, were measured in the Earth's field (H = 0.05 MT) with a Cenco MS-3B susceptibility bridge unit. Stepwise demagnetization experiments were carried out using: (1) an alternating field (AF) demagnetizer built at University Laval (Seguin, 1975) and a Schonstedt apparatus (Model GSD-1) in residual fields of less than 8 nT, and (2) a Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT. For the NRM of each oriented sample, a minimum of two and a maximum of four specimens (2.5 cm in diameter; 2.2 cm in height) were used to obtain a sample-mean direction. The sample-mean NRM direction (Fig. 3) is characterized by a relatively wide scatter of mainly positively inclined steep remanence and has a pronounced deviation away from the present Earth's field (PEF) at the site localities. Most of the NRM directions are present in two subvertical planes oriented respectively SE–NW and NE–SW (Fig. 3). These directions indicate that the basaltic flows do carry ancient remanence; indeed, the directions of the NRM for the samples located close to the PEF moved away from the initial directions during subsequent AF or thermal demagnetizations. The intensities of NRM (Jn) vary from 10^{-4} to 2 A m^{-1} with Koenigsberger ratios (Qn = Jn/kH) between 0.01 and 20. The predominant ratios are 5×10^{-2} for HA samples and 0.7 for HB samples (Fig. 3). It is clear that the Koenigsberger ratio is characteristic of the type of flows (HA or HB). The susceptibilities and Qn values are consequently reflecting the magnetic mineralogy of the two types of flows. NRM intensities of HA samples are usually smaller than those of the HB samples.

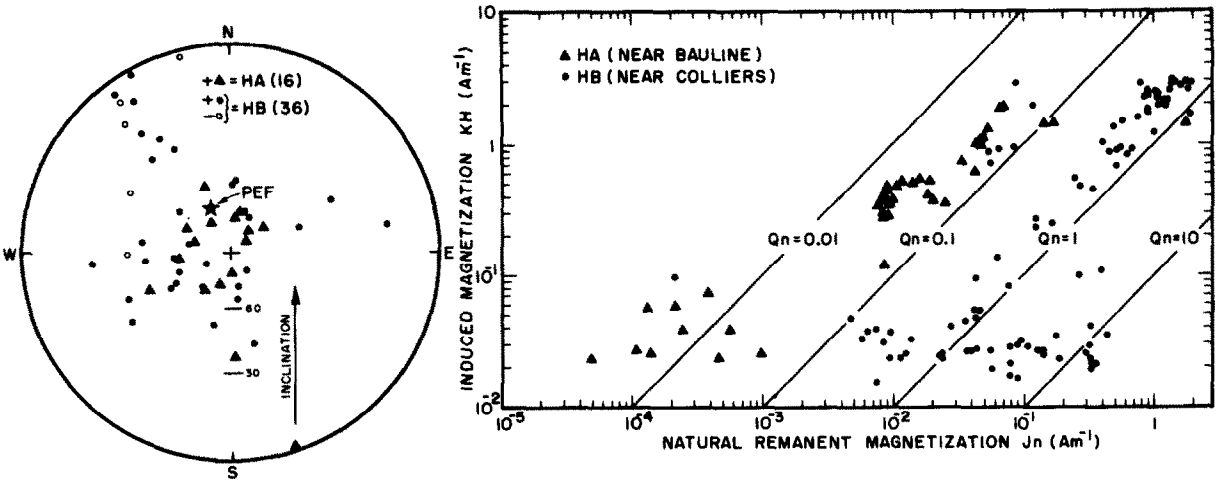


Fig. 3. Individual sample NRM directions on Wulff net for sites HB1–7, HA6–7, and HA15 (left). The solid (open) circles represent projections on lower (upper) hemisphere. The star represents the present Earth's field (PEF) at the sampling localities. Plots of intensity of NRM (Am^{-1}) against the induced magnetization (KH in Am^{-1}) with resulting slope lines indicating the values of Koenigsberger ratio Q_n (right) for all samples.

Paleomagnetic study

(a) AF demagnetization

A least one specimen from each of the 52 samples were subjected to AF demagnetization in steps of 5 mT in the range 5–30 mT and in steps of 10 mT in the range 30–100 mT. The demagnetization results were subsequently analysed using Zijderveld orthogonal plots (Zijderveld, 1967). Although an increase in the precision parameter k of the mean direction after AF demagnetization is not always noted, there is a shift of the mean direction away from the PEF at all sites, indicating either complete or partial removal of minor viscous remanence. For both within and between

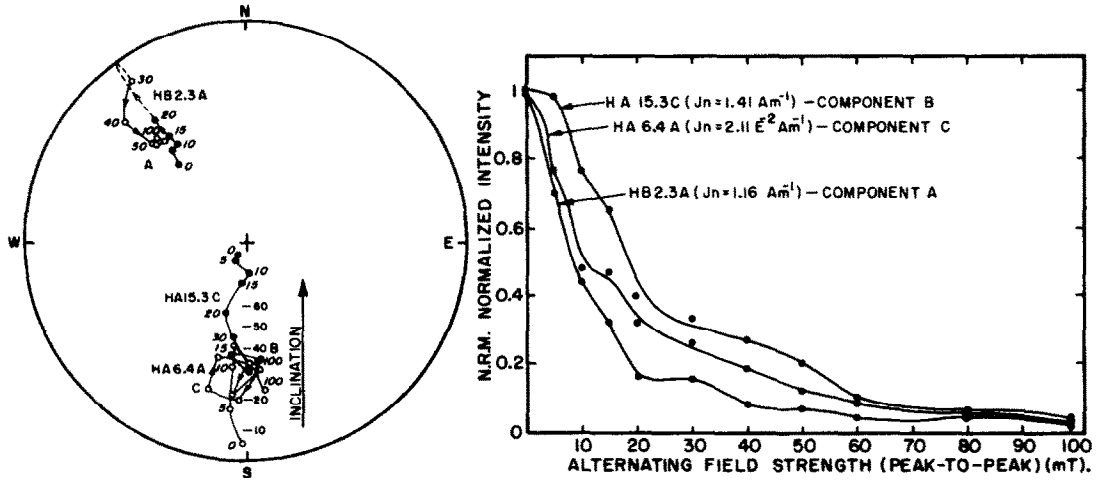


Fig. 4. Equal-area projections of directions of magnetization (left) of three specimens from different metabasalt samples after AF treatment. The numbers next to solid or open circles represent the demagnetization steps in mT as indicated. Solid (open) circles represent projection on the lower (upper) hemisphere. The respective normalized AF decay curves are shown on the right side. This figure illustrates an example for the isolation of the three components (A, B and C) of magnetization.

site locations, the agreement in directions was much improved after the removal of the viscous component in a field of 5–20 mT. The specimens lost 50% of their NRM intensity in the 10–20 mT range and more than 90% in the 30–40 mT range for samples with a magnetite memory carrier (Fig. 4). In this case, the residual component is best isolated in the 15–40 mT range. However, more samples contain both magnetic and hematite memory carriers than only magnetite or only hematite. In this last case, the stable component is best extracted in the 610°–640°C and 50–100 mT ranges. A few samples (e.g. in sites HA-6, HA-7, HB-1) are instable upon AF demagnetization; this represents 10–15% of the population studied.

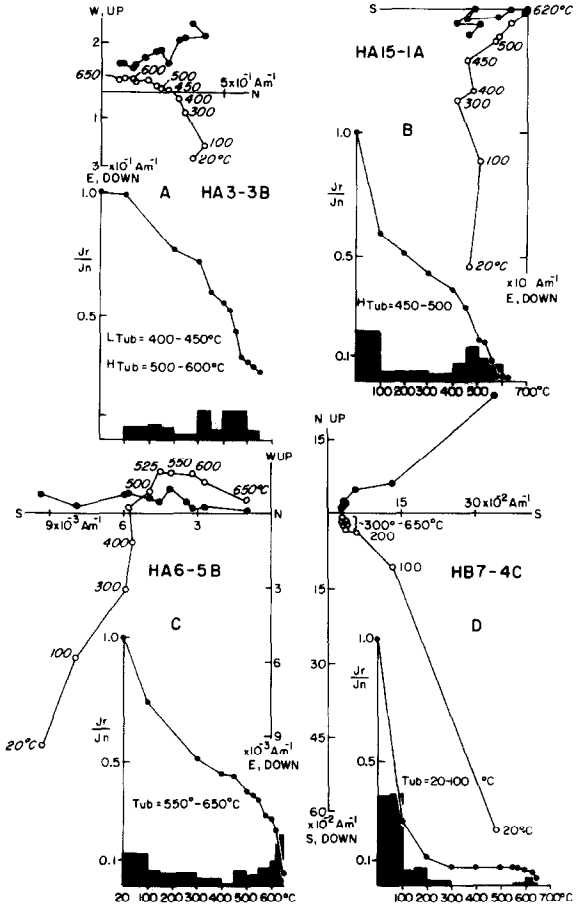


Fig. 5. Representative orthogonal thermal demagnetization diagram (Zijderveld, 1967) of successive demagnetization vectors for four specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. The numbers next to the circles represent the demagnetization steps in °C as indicated. The unblocking temperature (T_{UB}) in °C is indicated below the orthogonal plots. This figure illustrates an example for the isolation of the four components (A, B, C and D) of magnetization.

(b) Thermal demagnetization

Approximately 60% of the specimens were thermally demagnetized. In general, thermal demagnetization is more efficient at reducing secondary components than AF demagnetization in view of the important content of hematite in the aerial or subaerial volcanic units. For the samples in which magnetite is the main memory carrier, the stable component is isolated in the 450°–575°C range (Fig. 5). Where hematite is predominant, the stable component is extracted in the 600°–650°C range. Approximately 60–70% of the samples have a predominant hematite memory carrier. Magnetite and hematite were identified with the aid of reflected light microscopy in one representative sample per site.

Components isolated

Four (4) components of magnetization (A, B, C, D) were isolated by both AF and thermal demagnetizations. Component A has a NW orientation with shallow negative inclination, component B a SSE orientation with shallow to intermediate inclination, component C a SSW orientation with shallow negative inclination and component D a W orientation with intermediate negative inclination (Table 1). Component A is characterized by a normal (N) polarity, component B by a reverse (R) polarity and components C and D by mixed (M) polarities that are predominantly reverse. Generally, only one component is isolated in a single sample but often two or three components are present in a given site. Site HB-2 is an exception; it contains component A only. Vector trajectories are usually linear until stable endpoints are reached at high fields or temperatures and univectorial directions of magnetization are obtained once the low coercivity or low temperature soft component is removed. This soft compo-

nent is observed in approximately 40% of the samples. Specimens HB2.3A and HB3.3B are representative examples of the behaviour of the A(N) components (Figs. 4 and 5a), specimens A15.3C and HA15.1A for the behaviour of the B(R) component (Figs. 4 and 5b), specimens HA6.4A and HA6.5B for the behaviour of the C(M) component (Figs. 4 and 5c) and specimen HB7.4C for the behaviour of the D(M) component (Fig. 5d).

Component A is encountered in seven of the ten sites; its unblocking temperature (TUB) and median destructive field (MDF) range between 550° and 650°C, 1 and 100 mT respectively and its memory carrier is mainly hematite. Component B is found in seven sites also; its T_{UB} and MDF vary from 450° to 585°C and 10 to 30 mT and the memory carrier is magnetite.

For component C, the T_{UB} and MDF on seven sites range from 525°C to 600°C and 20 to 30 mT respectively, the magnetic memory carrier is predominantly magnetite with some hematite. Component D is encountered on two sites only (HB-1 and HB-6); some samples from these sites (particularly HB-6) are largely overprinted by the PEF. This last component is either unstable or insufficiently cleaned and for these reasons it will not be retained in the calculation of paleopole positions.

Statistical analysis and nature of magnetization

Site-mean directions for AF and thermally demagnetized results combined are shown in Table 1 and Fig. 6. The A component is the most stable and is followed by components B and C. What then is the age of magnetization and the sequence of acquisition of these components? To answer these questions, the tilt-correction test and reflected light examination of the opaque minerals are useful. The ratio of the precision index (McElhinny, 1964) shows that a significantly negative tilt test is observed in components A and C. Microscopic observations reveal that hematite is secondary after magnetite. Even if the tilt test is significantly negative, the frequency distribution (number of specimens within a component), T_{UB} and MDF are larger for A than for B and C and this may be an indication that component A is

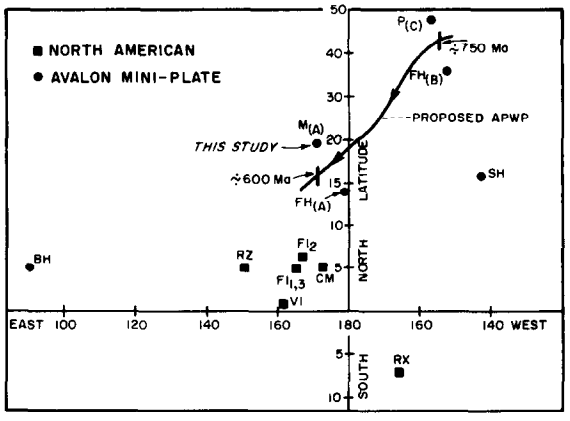


Fig. 6. Paleomagnetic poles (Late Hadrynian) derived from the Avalon zone and cratonic North America. Poles RX and RZ are from Morris (1977) and poles $FI_{1,2,3}$ and VI from Palmer et al. (1983).

older than components B and C. Component B which does not pass the tilt test may be younger or older than component C. The frequency of components B and C is the same but the T_{UB} and MDF of component C are slightly higher than those of component B. Sometimes, component C is associated with low field PEF component and this suggests that this component is more unstable than component B. In addition, the samples containing component C are usually more altered. For these reasons, component A is considered as the oldest of all the secondary magnetic components. Component B which is presumably of Paleozoic age has a mean orientation comparable to those of FG and G granitic units of Devonian age from Ste-Cécile-St-Sébastien Region, Quebec Appalachians (Seguin et al., 1982) and the St.Lawrence granite N magnetization (Irving and Strong, 1985); it is most probably of upper Devonian age corresponding to the Acadian orogenic event. The mean direction of magnetization for component C is characteristic of remagnetizations corresponding to Permo-Carboniferous overprints (Roy and Morris, 1983; Irving and Strong, 1985). Component D is interpreted as an unresolved component as its spectrum overlaps another component. A similar pseudo-component was obtained in lower Devonian volcanic units from New Brunswick and it was also explained as an overprint (Seguin and Gahé, 1985).

Component A ($D = 317$, $I = -22$) appears to

TABLE 1

Mean directions of cleaned components for basaltic and mafic flows: Avalon peninsula, Newfoundland *

Site No:	Strike (°)	Dip (°)	Component A						Component B					
			N	*n*	*D*(°)	*I*(°) [a]	α_{95}(°)	*k*	*N*	*n*	*D*(°)	*I*(°)	α_{95}(°)	*k*
In-situ														
HB-1	2	65W							5	13	186	37	26.2	9.5
HB-2	350	68W	5	16	320	−24	21.1	14						
HB-3	355	75W	5	12	315	−15	9.1	72	1	2	160	40	–	–
HB-4	352	70W	2	4	312	−31	–	–	2	3	163	27	–	–
HB-5	345	72W	3	8	321	−10	22.6	31	2	4	191	21	–	–
HB-6	314	68W	2	5	314	−26	–	–						
HB-7	339	66W												
HA-6	17	84E(O)	3	5	314	−27	28.3	20	2	4	169	34	–	–
HA-7	48	75 NW							4	6	179	46	27.6	12
HA-15	14	74W	2	3	325	−20	–	–	4	7	177	29	9	105
Total			7		317	−22	6.3	92	7		175	34	9.8	39
Tilt-corrected														
HB-1											217	12	P:N	
HB-2					005	−35	P:N							
HB-3					002	−42	N				213	21	N	
HB-4					006	−44	N				195	17	N	
HB-5					348	−26	N				195	−17	N	
HB-6					314	−10	N							
HB-7														
HA-6					069	−49	N				236	19	N	
HA-7											270	40	N	
HA-15					026	−51	N				219	22	N	
Total					004	−41	23.8	8			219	18	22.5	8.2
					$k_1/k_2 = 11.5$						$k_1/k_2 = 4.8$			

Site No:	Strike (°)	Dip (°)	Component C						Component D				
			N	*n*	*D*(°)	*I*(°)	α_{95}(°)	*k* [a]	*N*	*n*	*D*(°)	*I*(°)	α_{95}(°)
In-situ													
HB-1	2	65W	1	2	201	−27	–	–	3	5	271	−41	31
HB-2	350	68W											
HB-3	355	75W											
HB-4	352	70W			187	−16	–	–					
HB-5	345	72W	1	3	176	−23	25.4	26					
HB-6	314	68W	3	9	201	−45	–	–	4	11	262	−33	19.2
HB-7	339	66W	1	5	175	−37	37	12					
HA-6	17	84E(0)	3	6	180	−19	38	7					
HA-7	48	75NW	4	8	208	−21	28.4	20					
HA-15	14	74W	3	6	190	−27	12	26	2	4	266	−37	–
			7										
Tilt corrected													
HB-1					163	−27	P:N				085	−74	P:M
HB-2													
HB-3													
Hb-4					155	−25	M						
HB-5					123	−46	M						
HB-6					118	−46	R				020	−75	M
HB-7					159	21	N						
HA-6					188	32	M						
HA-7					177	−18	M						
HA-15					157	−18	31.1	4.7			054	−77	
Total					$k_1/k_2 = 5.5$								

* *N*, *n*, 0 = number of samples. specimens; overturned; P, N, R, M = polarity, normal, reversed, mixed.

[a] All significantly negative at 95% confidence level.

be older than components B and C is pre-upper Devonian and syn- or post-Vendian (600 Ma) in age. No specific age of acquisition can be tagged to this component but a somewhat similar orientation of magnetization ($D = 308$, $I = -25$) was isolated in the Finn Hill ignimbritic sequence in the Avalon zone of eastern Newfoundland (Seguin, 1985). It is possible that component A is an unresolved pseudo-component but since the NW direction is characterized by a shallow negative inclination, this suggests an old age of magnetization (i.e. Cambrian or very Late Precambrian). For this reason, component A is tentatively related to the Cadomian orogeny.

Paleopole positions and their relation to poles previously determined

Component A ($D = 317°$, $I = -22°$) yields a paleopole at 20°N, 172°E; component B ($D = 175°$, $I = 34°$) a paleopole at 24°N, 132°E and component C ($D = 190°$, I $= -27°$) a paleopole at 56°N, 109°E (see Table 1). North American paleomagnetic data for the time interval: 800–550 Ma are scarce (Fig. 6). For this same age span, only five paleopoles (BH, FH(A), SH, FH(B), P(C)) are available on the Avalon microcontinent (Fig. 6). In the section dealing with sequence of acquisition of magnetization, it was concluded that component C is secondary, i.e. Late Paleozoic or possibly Mesozoic. The paleopole of component C is not located in the proximity of Early Paleozoic poles from cratonic North America or northeastern Europe after closure of the present Atlantic Ocean but is near Early Permian poles, indicating that the metabasalts of the Harbour Main Group were remagnetized in the Late Paleozoic or slightly later. Component B appears to be younger than component A according to the conclusions arrived at in the previous section. Nevertheless, the latitude of pole B is lower than that of pole C. This suggests a somewhat older age. According to an updated version of the Devonian APWP for North America (Seguin, 1986), a Late Devonian age characterizes this pole position. Assuming that this interpretation is correct, component B represents the Acadian orogenic event. The paleopole of components A has a latitude somewhat lower than that of component B. The paleopole of this component (20°N, 172°E) is close to that of FH(A) (14°N, 179°E) which was interpreted as the magnetic imprint of the Avalonian (Cadomian) orogenic event. The nearby Early Cambrian Brigus shale located unconformably above the Late Precambrian metabasalts is characterized by westerly intermediate positive direction which is definitely unrelated to the direction of magnetization of component A in this study. The paleopoles obtained from the Cloud Mountain basalt (Deutsch and Rao, 1977) and the Black Head Formation (BH) and Signal Hill (SR) Group (Nairn et al., 1959), components $H_{(A)}$ and $H_{(B)}$ of the Finn Hill Sequence (Seguin, 1985), component $P_{(C)}$ of the porphyrites (Seguin and Lortie, 1986) are 5°S, 8°W; 5°S, 96°W; 16°S, 38°E; 14°S, 1°W; 39°S, 29°E and 48°S, 31°E (Fig. 6). The first rock unit does not belong to the Avalon zone. The inclusion of the SH and BH poles in a preliminary Late Precambrian APW path for Avalon is unjustified because of the poor reliability of these poles (E.R. Deutsch, pers. commun., 1986). Thus, only the Finn Hill sequence, the porphyrites and rhyolites pole as well as pole of component A in this study can be used to construct a suitable APWP for the Avalon zone. Pole $FH_{(B)}$ (Seguin, 1985) and pole $P_{(C)}$ (Seguin and Lortie, 1986) are characterized by a primary component while poles $FH_{(A)}$ and $M_{(A)}$ (this study) are characterized by a secondary component. If the interpretation of the relative chronology of the paleopoles is correct, then a specific trend for the APW trajectory is apparent (see proposed APWP of Fig. 6). From approximately 750 to 600 Ma, the APW path moves westward and southward from 45°N, 150°W to 15°N, 175°E.

What does the present paleomagnetic study tell us about the relative displacement of Avalon and other Atlantic bordering terranes? Let us start with a comparison of the NA craton and the Avalon Zone. According to the existing paleomagnetic data for the two tectonic units, the significant difference between Avalon pole positions and contemporary pole position from the North American craton implies the presence of two separate plates around 650–700 Ma and older ages. It

is worth noting that at approximately 600 Ma, the paleopoles $FH_{(A)}$ and $M_{(A)}$ from the Avalon microplate are not significantly different from the paleopole obtained in the Cloud Mountain basalt overlying the Grenville basement of northwestern Newfoundland. This suggests that in Late Precambrian time, the eastern and western Newfoundland blocks were not very distant from each other or else they were far apart but at the same paleolatitude. Assuming that the two blocks were not at the same paleolatitude, they were positioned 500–1000 km away and a pre-Iapetus small ocean probably existed. As the Precambrian units of western Newfoundland were not affected by the Cadomian orogeny positioned in the interval of 690–550 Ma, it implies that eastern Newfoundland never came in contact with the North American craton in Late Precambrian time. Using geological constraints, many hypotheses of possible relative displacements of these two terranes were advanced, but two are more acceptable. Hypothesis 1—Ziegler et al. (1977) stated that the Avalon plate was a strike-slip microcontinent oriented more or less parallel to the eastern edge of the North American craton. Hypothesis 2—According to Rast and Skehan (1983), the Avalon plate consisted originally of an island arc at the edge of the Cadomian orogen in compression between the west African craton and the Laurentian continent in Late Precambrian time. In the first hypothesis, an intermediate to large paleolatitude discrepancy should be observed in paleomagnetic data originating from the two different blocks while in the second hypothesis, this needs not be the case. Consequently, the second hypothesis appears more probable on paleomagnetic grounds.

Next, how does the Avalon zone relate to other Atlantic-bordering terranes such as northwestern Europe and North Africa and what is the nature of their respective relative motions?

According to Van der Voo (1982), the paleomagnetic information with regard to North America (Laurentian) plate is sparse for the Late Proterozoic and it is abundant for the African plate. A closer look at the paleomagnetic literature from the Proterozoic of Africa (Brock, 1981) indicates that the amount of Proterozoic paleomagnetic data from Africa is also scarce. A comparison of the Late Proterozoic apparent polar wander path (APWP) of Africa and northwestern Europe (Briden and Duff, 1981) shows some similarities. For the 600–800 Ma interval (Kröner et al., 1980), the APWP for Africa is represented by a V-shaped swath pointing to the south and centered on longitude 270° between latitudes: 30°–70°S. The APWP of northwestern Europe for the same time interval is still ill-defined (Hagstrum et al., 1981; Perigo et al., 1983) with one cluster of poles at 120°E, 37°N and another at 230°E, 30°N. The APWP for northwest Europe defined by Briden and Duff (1981) is also represented by a V-shaped trajectory pointing to the east and centered at latitude 30°N between latitudes 230° and 275°. During this time interval, the shape of the swath and the angular excursion (45°) is the same for North Africa and northwestern Europe. This similarity in path geometry suggests that there was no relative motion of North Africa and northwestern Europe for this time interval and that these two blocks were probably part of a same plate during Late Proterozoic time. Post-Precambrian rotations are responsible for the actually differing APWP positions.

The Proterozoic APWP of the Avalon zone is different from that of North Africa and northwestern Europe. As defined today, it is not characterized by the V-shaped swath. This dissimilarity in paleomagnetic signature suggests that the Avalon microcontinent is unrelated either in origin and/or motion to the North Africa–northwestern Europe plate in Late Proterozoic time. A possible scenario of plate tectonics motion of Avalon in Hadrynian time is discussed in Séguin (1985). The significant differences in this investigation are the detection of the superimposed Permo-Triassic and Acadian orogenic events. Obviously, much more paleomagnetic data are needed to come up with a clearer picture of the Late Proterozoic APWP of Avalon. In spite of this shortage of data, we believe that the line of reasoning relative to plate tectonics should be maintained in future studies of the Avalon zone.

Conclusion

Three components of magnetization (A, B and C), all of which are secondary, were isolated in

late Proterozoic basalts of the Avalon terrane of eastern Newfoundland. The paleopole position of component C suggests a Permian or younger age of acquisition. The paleopole position of component B is characteristic of a Middle–Upper Devonian age (timing of the Acadian orogeny). Component A is also an overprint but it is the oldest component; its paleopole position is different from that of the Paleozoic poles of the Avalon zone. It is interpreted as the magnetic signature of the Late Proterozoic Cadomian orogeny. This last pole is correlated with the one obtained in a previous paleomagnetic study from the Avalon of Newfoundland. Combining the results of this study with those from previous investigations, a preliminary APWP of the Late Proterozoic of Avalon is presented. An attempt was made to calculate the relative displacement of Avalon with other Atlantic bordering terranes (cratonic North America (NA), northwestern Africa (AF) and northwestern Europe (E)). In the 650–700 Ma or older time interval, cratonic NA and Avalon were two widely separated and distinct plates. At approximately 600 Ma, they were probably closer to each other. It is pointed out that the APWP of Africa and northwestern Europe shows peculiar similarities during Late Proterozoic time. The dissimilarity in paleomagnetic signature of AF and E on the one hand and Avalon on the other hand in Hadrynian time indicates that the Avalon microplate is unrelated in origin and/or motion to the north AF and northwestern E plate during most of the Late Proterozoic timespan.

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